

Heng Diao

M.A. Candidate in Applied Linguistics

Tianjin University of Finance and Economics

Corpus Linguistics | Computational Linguistics | Research Software Development

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Research Portfolio

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PROFILE

I am interested in using computational methods to solve linguistic research problems. I have developed research software for large-scale corpus analysis and linguistic annotation, including tools implementing Biber's Multi-Dimensional Analysis and his corpus-based framework for stance analysis. I am open to developments in AI and interested in exploring how they can be applied to linguistic research and used to address broader social needs.

RESEARCH INTERESTS

Corpus Linguistics; Computational Linguistics; NLP for Linguistic Research; Stance and Evaluation; Register Analysis; Research Software Development; Language Technology

EDUCATION

Tianjin University of Finance and Economics, Tianjin, China

Sep. 2024 – Jun. 2027 (expected)

M.A. in Applied Linguistics

GPA: 3.92/4.00

Master's Thesis:

Stance Expressions in Signed Commentaries on China's Economy: A Deep Learning-Based Corpus Analysis.

Henan University of Technology, Zhengzhou, China

Sep. 2017 – Jun. 2022

B.A. in English

PATENTS

- Liu, T., & Diao, H. (2026).
A Method and System for Stance Annotation of English Texts.
Chinese Invention Patent, Patent No. ZL202610057958.7.
Rule-based stance annotation and model training based on Biber's stance framework.
- Liu, T., & Diao, H. (2026).
An Intelligent Analysis and Text Optimization Method and System for External Financial Discourse.
Chinese Invention Patent, Patent No. ZL202610071079.X.
Quantitative analysis of financial discourse integrating Multi-Dimensional Analysis and stance analysis.

RESEARCH SOFTWARE

DDU Linguistics Lab — Integrated Corpus Research & Analysis Platform

2024 – Present

Research Software Developer

- Develop a research platform that translates corpus-linguistic research questions into software tools for corpus retrieval, annotation, statistical analysis, and visualization.
- Develop three core products: *Multi-Dimensional Analysis*, *Stance Analysis Tagger*, and *Discourse Analysis System* (ongoing).
- Build backend services and data infrastructure using Python, SQLite/PostgreSQL, Redis, and NLP pipelines such as Stanza.
- Use AI coding tools such as Codex and Claude Code throughout the development workflow.

Corpus Annotation Framework

2025 – Present

Annotation, Dataset Management, and Model Training

- Design a collaborative annotation process involving task assignment, span-level annotation, cross-verification between annotators, and final adjudication by experts.
- Integrate collaborative annotation, dataset management, and model training, allowing verified annotations to be used directly for training.
- Add an experimental pre-annotation feature using multiple AI agents, with human review before inclusion in the validated dataset.

Terminology Management and Bilingual Translation Platform
Computer-Assisted Translation Platform

2026 – Present

- Develop a computer-assisted translation platform for bilingual text alignment, terminology management, collaborative translation, and editing.
- Build a terminology database for term storage, retrieval, annotation, and consistency checking.
- Build and manage aligned bilingual corpora to support terminology and corpus-based translation research.

RESEARCH OUTPUTS

Publication

- Diao, H., & Liu, T. (2025).
Research Tool Development and Application for External Communication News Corpora in the Digital Intelligence Era.
In *2025 Collection of Key Academic Achievements in Tianjin Social Sciences*.
Tianjin People’s Publishing House.

Conference Presentation

- Diao, H. (2025).
Building a Visual Corpus Platform Based on AI-Assisted Programming and Python.
Paper presented at the 11th Annual Conference of Tianjin Foreign Language Society, “Opportunities and Challenges of Foreign Language Education in the AI Era,” Tianjin, China.

Manuscript in Preparation

- Diao, H. (2026).
The Discursive Stance of LLMs on China’s Economic Issues: A Corpus-Based Analysis of AI-Generated Texts.
Manuscript in preparation.

AWARDS

- First-Class Graduate Academic Scholarship, Tianjin University of Finance and Economics (awarded twice).
- Outstanding Paper Award, Tianjin “New Era Young Scholars Forum” (2025).
- Second Prize, Graduate Division, Tianjin Region, First National Artificial Intelligence Application Innovation Competition (2025).

TECHNICAL SKILLS

Research Methods	Data cleaning; corpus construction; retrieval optimization; visualization
NLP & AI	Stanza; sequence labeling; model training; agentic workflows; Ollama
Development	Python; SQLite; PostgreSQL; Redis; Docker; JavaScript; HTML & CSS

LANGUAGES

Chinese	Native
English	Advanced proficiency